Where Shoppers Drop Off

E-commerce funnel, product and retention analytics in SQL

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Analyst's technical report. SQL project (SQLite dialect) for a Product Manager's questions on engagement, conversion, abandonment, product performance and retention.

Data: synthetic e-commerce event database (ecommerce_clean.sqlite) · 26 weeks, 7 Jul 2025 - 4 Jan 2026 · 30,000 users · 89,557 sessions · 590,648 events · 6,896 orders

Item	Status
sql/ecommerce_product_analytics.sql (one file, 7 sections)	2,072 lines. Executed end to end with the sqlite3 shell 3.45.1 (exit code 0, 33.4 s) and with SQLite 3.53 through the Python runner (70 blocks, 35.5 s). The two runs produced identical contents in all 30 tables and views.
Validation checks (Section 7)	27 of 27 PASS, 0 FAIL. A seeded-error test on a copy made 6 checks fail as expected.
Raw data	Attached read-only; MD5 451c337ac01d6286b2a1575b246ed178 before and after every run.
Result files	41 CSV outputs (34 analytical queries Q01-Q28, 5 data-quality and 2 validation outputs) and 7 cleaned-table exports.

1. Project overview

Item	Detail
Project title	Where Shoppers Drop Off: E-commerce Funnel, Product and Retention Analytics in SQL
Business scenario	A Product Data Analyst at an e-commerce technology company. The Product Manager wants to understand how users interact with the platform, where they abandon their shopping journeys, and which products or user behaviours are associated with successful purchases. The task is to turn the event records into product performance metrics.
Business problem	The company sees traffic and orders, but not the journey between them. Without a measured funnel it cannot tell which stage to fix, which products waste attention, or which users and channels bring lasting engagement.
Purpose of the analysis	Measure engagement, build the view-to-purchase funnel, locate abandonment, assess products and categories, measure returning-user behaviour and cohort retention, and turn the results into prioritised opportunities.
Scope	All 590,648 events, 89,557 sessions, 30,000 users, 6,896 orders and 250 products in the 26-week window. Descriptive and diagnostic. Out of scope: causal attribution (no experiments in the data), forecasting, profitability and channel return on spend (no cost data).
Tool used	SQL, SQLite dialect (version 3.35 or later required; tested on 3.45.1 and 3.53). The whole analysis is one SQL file. Python is used only to run that SQL and export results (standard library) and to draw charts from the exported results (pandas, matplotlib). No analysis is done in Python.

2. Business objectives and analytical questions

2.1 Objectives (as supplied)

1. Measure user engagement with the digital platform.
2. Analyse the conversion funnel from product views to purchases.
3. Identify stages at which users abandon their shopping journeys.
4. Examine product and category performance.
5. Analyse returning user behaviour and retention patterns.
6. Identify opportunities to improve conversion and user engagement.

2.2 Analytical questions (as supplied) and how they were translated

Question	Analytical requirement	Output
How many unique users interacted with the platform?	Distinct users with a session; registered vs guest	Q01
How many product views were recorded?	Count product_view events, and separately those traceable to a product	Q01
How frequently do users return to the platform?	Sessions per user (distribution), gap between sessions, weekly returning users	Q03, Q03b, Q04
What is the average number of product interactions per session?	Define product interaction; mean and median per session	Q02
Which products receive the most views?	Rank products by views; view concentration	Q06, Q06b
How many users viewed products, added items to their carts and completed purchases?	Closed user funnel with purchases from orders (system of record)	Q07, Q07b
What percentage of viewers add products to their carts?	Step conversion view → cart	Q07, Q08
What percentage of cart users complete a purchase?	Cart → checkout → order	Q07, Q08
At which stage does the largest drop-off occur?	Rank step drop-offs; by platform, device and source; behaviour after add-to-cart	Q07, Q09–Q12b
Which products have the most views?	As above, with conversion alongside	Q06
Which products have the highest view-to-purchase conversion?	Viewer → buyer rate per product, ranked by Wilson lower bound (≥ 500 viewers)	Q14
Do popular products necessarily generate the most purchases?	Spearman correlation; overlap of top lists; popularity / conversion quadrants	Q13, Q13b, Q26
Which categories have the highest purchase activity?	Orders, buyers, units, conversion and shares by category	Q16
Which products experience relatively high cart abandonment?	7-day cart outcome, censoring-aware; product vs category and overall	Q17, Q17b
How many users return after their first observed interaction?	Return on a later day; within 7 / 14 / 28 days among eligible users	Q19

Question	Analytical requirement	Output
How does activity change across weekly user cohorts?	Cohort × week-offset retention matrix (recursive CTE)	Q20, Q22
What proportion of users return after one, two or four weeks?	Pooled week-1 / 2 / 4 retention over observed cohorts; three populations	Q21
Which user groups show stronger repeat engagement?	Return and repeat-buying by channel, device, city, registration, loyalty, first-session outcome	Q23, Q24

Objective 6 draws on all of the above. Q25 consolidates the sized opportunities, and Q18 adds promotion periods. The full objective-to-query mapping, with techniques, is in docs/project_brief.md.

3. Dataset overview

Source file: ecommerce_clean.sqlite (66 MB), supplied with the brief and stored unchanged at data/raw/. The data is **synthetic**. It simulates a Nigerian marketplace (the app user agent is "OjaLoop", the brands are fictional, the cities are Nigerian). It does not describe a real business.

Table	Rows	Grain	Key fields
events	590,648	one interaction	event_type, product_id, session_id, user_id, order_id, event_timestamp
sessions	89,557	one visit	device_type, platform, traffic_source, session_start/end
users	30,000	one user	signup_date (NULL = guest), acquisition_channel, city, is_loyalty_member
orders	6,896	one order	order_status, items_subtotal, session_id
order_items	10,808	one order line	product_id, quantity, unit prices, line_total
products	250	one product	category_id, list_price, is_active
categories	10	one category	category_name, department
promotions / promotion_products	6 / 311	promotion; product in promotion	start/end dates, scope, discount_pct
calendar	182	one day	week_index (1–26), is_weekend, promotion_name

3.1 Structure

The data is a star-like transactional model. events is the fact table at the finest grain and points to sessions, users, products and (for purchases) orders. orders points to users and to the session in which the order was placed, and order_items holds its product lines. The calendar provides the Monday-start weeks used for every weekly metric and cohort. Section 2 of the SQL recreates every relationship as a declared FOREIGN KEY, and PRAGMA foreign_key_check returns no violations.

3.2 Event types

event_type	Events	% missing product_id	% missing order_id
product_view	311,955	0.45%	100.00%
page_view	157,849	100.00%	100.00%
search	65,586	100.00%	100.00%
add_to_cart	28,941	0.44%	100.00%
purchase	10,497	0.00%	0.00%
begin_checkout	9,816	100.00%	100.00%
remove_from_cart	6,004	0.00%	100.00%

3.3 Dataset limitations

- The data is synthetic, so the findings demonstrate method rather than facts about a real company.
- Registration, loyalty and acquisition channel are recorded at extract date, not at the time of each event.
- The currency is not stated (₦ is assumed). There are no cost, margin or marketing-spend fields, and no page-level or checkout-step detail.
- The window is 26 weeks, so users active before it look new at the start (left-censoring), and late users have little follow-up (right-censoring).

4. Data quality assessment

Data quality was assessed in SQL (Section 3 of the script). Every check writes one row to a dq_register table: the dimension, the affected rows, the total, and the action taken. Loading through constraints (Section 2) is a check in itself: a duplicate key, orphan reference or invalid value would stop the load. None did.

4.1 Load audit

Table	Source rows	Loaded rows	Status
events	590,648	590,648	PASS
sessions	89,557	89,557	PASS
users	30,000	30,000	PASS
order_items	10,808	10,808	PASS
orders	6,896	6,896	PASS
promotion_products	311	311	PASS
products	250	250	PASS
calendar	182	182	PASS
categories	10	10	PASS
promotions	6	6	PASS

4.2 Data-quality register

ID	Dimension	Check	Affected	%
DQ01	completeness	product_view events with no product_id (untraceable product)	1,414	0.45
DQ02	completeness	add_to_cart events with no product_id	127	0.44
DQ03	completeness	traceable product_view events with no displayed price	901	0.29
DQ04	completeness	users with no signup_date (guest visitors who never registered)	19,414	64.71
DQ05	uniqueness	duplicate events (same session, timestamp, event type and product)	0	0.00
DQ06	integrity	foreign-key violations reported by PRAGMA foreign_key_check	0	0.00
DQ07	consistency	events whose user differs from the session user, or that fall outside the session window	0	0.00
DQ08	consistency	orders with no purchase event in the event stream (tracking loss)	206	2.99
DQ09	consistency	order lines with no matching purchase event	311	2.88
DQ10	consistency	orders whose subtotal or discount does not equal the sum of their lines	0	0.00

ID	Dimension	Check	Affected	%
DQ11	consistency	orders placed in a session with no add_to_cart event (cart filled in an earlier session)	1,934	28.05
DQ12	validity	product events dated after the product's discontinued_date	10	0.00
DQ13	validity	product views priced below list price with no promotion active for the product	2	0.00
DQ14	validity	sessions with session_end = session_start (single-event sessions)	392	0.44
DQ15	timeliness	registered users whose signup_date is after their first observed session	1,959	18.51
DQ16	timeliness	sessions ending after the last calendar date (crossing midnight on 4 Jan 2026)	1	0.00
DQ17	validity	sessions with a bot-like user agent (bot, spider, crawl, headless, curl, python)	0	0.00
DQ18	validity	products priced at >= 4x or <= 0.25x their category median list price	12	4.80
DQ19	validity	staged event values needing conversion: non-integer ids or amounts, non-ISO timestamps, untidy event types, blank ids	0	0.00

Shaded rows affect more than 100 records. The actions taken are in Section 5 and in docs/data_quality_and_cleaning_log.md.

4.3 What the assessment found

- **Tracking loss (DQ08, DQ09).** 206 orders (2.99%) and 311 order lines have no purchase event. Counting purchases from events would give 4,556 buyers instead of 4,655.
- **Cross-session journeys (DQ11).** 1,934 orders (28.05%) were placed in a session with no add-to-cart. The cart was filled in an earlier session. This is not an error, but it means a session-only funnel would misread these journeys.
- **Untraceable product events (DQ01, DQ02).** 1,414 product views and 127 cart additions have no product_id.
- **Implausible prices (DQ18).** 12 of 250 products are priced at ≥ 4× or ≤ 0.25× their category median. For example, a wireless mouse is listed at 241,000 and a mechanical keyboard at 565,000, while some inverters cost 13,800.
- **Guests (DQ04) and late registration (DQ15).** 64.71% of users never registered, and 1,959 registered users signed up after their first observed session.
- **No problems found** in uniqueness (DQ05), referential integrity (DQ06), event–session consistency (DQ07), order arithmetic (DQ10), bot traffic (DQ17) or staging formats (DQ19).

4.4 Distribution and outliers

Metric	Min	P25	Median	P75	P99	Max	Mean
events per session	1	2	5	9	29	82	6.6
sessions per user	1	1	1	3	19	81	2.99

Both distributions have long right tails. A user with 81 sessions or a session with 82 events is plausible behaviour rather than an error, and no bot signatures were found, so the tails were kept. Medians are reported alongside means.

5. Data cleaning and preparation

The guiding rule was **keep and flag, never delete**. Every record is retained, and problem records carry 0/1 flags, so each metric can state exactly what it includes. Cleaning and transformation are in Section 4 of the SQL file. They produce an analysis layer of derived tables that every later query reads.

5.1 Issues, decisions and reasons

Issue	Decision	Reason
Orders missing from the event stream (DQ08/09)	Purchases, buyers, units and revenue are counted from orders and order_items. Purchase events are used only for reconciliation (Q07b).	Orders are the system of record; event tracking loses 3%.
Cross-session journeys (DQ11)	The primary funnel is per user (closed funnel over the whole window). The session funnel is reported as a complement.	28% of orders would otherwise look like a cart abandonment followed by an unexplained purchase.
Product events with no product_id (DQ01/02)	Flag product_traceable. Included in engagement and funnel counts; excluded from product metrics.	They are real interactions, but they cannot be attributed to a product.
Guests (NULL signup_date, DQ04)	Recoded to is_registered = 0 and analysed as a segment.	NULL here means “not registered”, not “unknown”.
Users existing before the window	is_known_existing_customer = signed up before 7 Jul 2025 (6,000 users). Retention is also reported for cohorts 3–26 and for users not registered before the window.	Left-censoring inflates the retention of the first cohorts.
Limited follow-up for late users and carts	Return-within-N-days is NULL below N days of follow-up. Cohort cells exist only for observed weeks. Cart additions in the last 7 days are flagged is_censored.	Without this, late users and carts would be counted as lost (right-censoring bias).
Cancelled and returned orders	is_net_order flag. Net revenue = items_subtotal of net orders. Orders of every status count as purchase behaviour in the funnel.	The funnel measures intent; revenue measures value retained.
Implausible prices (DQ18)	Kept unchanged and documented. Product and category analysis relies on counts and conversion; revenue rankings are labelled indicative.	There is no reference price to correct against, and inventing one would be fabrication.
Single-event sessions, the session crossing midnight, discontinued-product views (DQ12–16)	Kept. Duration is 0 seconds for single-event sessions; sessions are dated by their start.	Too few to matter, or plausible behaviour.

5.2 Transformations and derived tables

Derived table	Rows	What it adds
events_enriched	590,648	Calendar week, weekend and promotion flags, category and list price, is_product_event, product_traceable, shown_discounted, and session_event_seq (ROW_NUMBER per session)
session_summary	89,557	Event counts by type (conditional SUM); has_* stage flags; order outcome and is_net_order; funnel_stage (CASE); user_session_seq and days_since_prev_session (ROW_NUMBER and LAG over a named WINDOW); duration (julianday)
user_summary	30,000	Registration flags, cohort week, follow-up days, first-session attributes, totals, ever_* funnel flags, days_to_return, censoring-aware returned_within_7d/14d/28d, is_repeat_buyer, engagement_segment
user_week_activity	57,734	One row per user per active week
cohort_retention	351	Cohort × week offset (0–25) with active users and retention rate. Offsets come from a recursive CTE; only observed weeks are kept
user_product_purchases	10,808	Order lines with buyer, timestamp, category and net flag (indexed for EXISTS lookups)
cart_add_outcomes	28,814	For each traceable add_to_cart: purchased_same_session, purchased_within_7d, removed_in_session, is_censored
product_performance	250	Views, viewers, cart adds and users, orders, units, buyers, viewer-buyers, net revenue, view-to-cart and view-to-purchase rates, Wilson lower bound, abandonment, RANK and DENSE_RANK
category_performance	10	The same metrics at category level, with window shares of views, orders and revenue

5.3 Exclusions

No rows were deleted. Exclusions apply only to specific metrics, and each is stated where it applies:

- untraceable product events are excluded from product metrics;
- censored users are excluded from return-within-N denominators;
- censored cart additions are excluded from abandonment;
- products below 500 viewers are excluded from the conversion ranking;
- products below 100 observed cart additions are excluded from the abandonment ranking.

6. Tool-specific execution (SQL)

6.1 How the script is organised and run

The work is one SQLite file, sql/ecommerce_product_analytics.sql (2,072 lines), in seven sections. Each executable block starts with a marker comment: -- @exec, -- @query or -- @export. These are ordinary SQL comments, so the file runs unchanged in the sqlite3 shell:

```
sqlite3 data/processed/ecommerce_analytics.db < sql/ecommerce_product_analytics.sql
```

scripts/run_sql_project.py executes the same file block by block. It writes each @query result to outputs/results/<name>.csv and each @export to data/processed/<name>.csv, logs timings, and stops at the first error. It contains no analysis.

Section	Content	Seconds
1 Setup	PRAGMAs, read-only ATTACH of the raw file, schema-qualified drops	0.0
2 Tables and loading	10 tables with PK/FK/UNIQUE/CHECK, events staging, 10 indexes, load audit, DETACH	5.5
3 Data quality	dq_register (19 checks), completeness, outlier profile	9.8
4 Cleaning and transformation	9 derived tables	9.5
5 Analysis	31 queries (Q01–Q25 including b-variants)	2.5
6 Views and outputs	8 views, Q26–Q28, 7 table exports	7.1
7 Validation	validation_results (27 checks)	1.1

6.2 Database setup, constraints and safe loading

What: the analysis database is created separately from the raw file. The raw file is attached in read-only mode, and every table is recreated with explicit types and constraints. **Why:** the raw data must never be overwritten, and constraints make the load itself a validation step. **How:**

```
ATTACH DATABASE 'file:data/raw/ecommerce_clean.sqlite?mode=ro' AS raw;
```

All drop statements are schema-qualified (DROP TABLE IF EXISTS main.events). The reason: SQLite resolves an unqualified name against attached databases when it does not exist in main. During development an unqualified drop tried to delete from the raw attachment, and the read-only mode stopped it. Qualifying every drop, and detaching the raw database straight after the load audit, makes the script safe to re-run.

Constraints encode the business rules of the source system. Some examples:

```
CREATE TABLE order_items (
    order_item_id    INTEGER PRIMARY KEY,
    order_id         TEXT    NOT NULL REFERENCES orders (order_id),
    product_id       TEXT    NOT NULL REFERENCES products (product_id),
    quantity         INTEGER NOT NULL CHECK (quantity > 0),
    unit_list_price  INTEGER NOT NULL CHECK (unit_list_price > 0),
    unit_sale_price  INTEGER NOT NULL CHECK (unit_sale_price > 0 AND unit_sale_price <=
unit_list_price),
    line_discount    INTEGER NOT NULL CHECK (line_discount >= 0),
    line_total       INTEGER NOT NULL,
    CHECK (line_total = quantity * unit_sale_price),
    CHECK (line_discount = quantity * (unit_list_price - unit_sale_price)),
```

```
    UNIQUE (order_id, product_id)
);
```

6.3 Staging and type conversion

Events, the table every metric depends on, pass through a TEMP staging table. Their format is profiled there, and the typed load applies explicit conversions, so that format drift in a future extract fails loudly rather than being coerced silently. The profile found 0 values needing conversion (DQ19).

```
INSERT INTO events (event_id, session_id, user_id, event_timestamp, event_type, product_id,
quantity, unit_price, order_id)
SELECT CAST(event_id AS INTEGER),
       TRIM(session_id),
       TRIM(user_id),
       strftime('%Y-%m-%d %H:%M:%S', event_timestamp),
       LOWER(TRIM(event_type)),
       NULLIF(TRIM(product_id), ''),
       CAST(quantity AS INTEGER),
       CAST(unit_price AS INTEGER),
       NULLIF(TRIM(order_id), '')
FROM stg_events;
```

6.4 Data-quality queries

Each check is an INSERT INTO dq_register SELECT …. Two techniques are worth noting. **DQ11** is an anti-join against a pre-aggregated CTE: a correlated NOT EXISTS on event_type made the planner scan the event_type index once per order, taking 150 s; the rewrite runs in under a second. **DQ18** computes a per-category median with window functions, because SQLite has no MEDIAN.

```
WITH cart_sessions AS (
    SELECT DISTINCT session_id FROM events WHERE event_type = 'add_to_cart'
)
INSERT INTO dq_register
SELECT 'DQ11', 'consistency', 'orders / events', 'orders placed in a session with no add_to_cart
event (cart filled in an earlier session)',
       COUNT(*), (SELECT COUNT(*) FROM orders),
       'Not an error: cross-session purchase journeys. The user-level funnel is the primary funnel; the
session funnel is reported alongside.'
FROM orders o
LEFT JOIN cart_sessions c ON c.session_id = o.session_id
WHERE c.session_id IS NULL;

WITH ranked AS (
    SELECT p.product_id, p.category_id, p.list_price,
           ROW_NUMBER() OVER (PARTITION BY p.category_id ORDER BY p.list_price) AS rn,
           COUNT(*)     OVER (PARTITION BY p.category_id)                      AS n
    FROM products p
), medians AS (
    SELECT category_id, MAX(CASE WHEN rn = (n + 1) / 2 THEN list_price END) AS
median_price FROM ranked GROUP BY category_id
)
…
```

6.5 Cleaning and transformation (Section 4)

Session sequence and gaps: ROW_NUMBER and LAG over a named window

```
), seq AS (
```

```
  SELECT
    s.*,
    ROW_NUMBER() OVER w                                    AS user_session_seq,
    LAG(s.session_start) OVER w                            AS prev_session_start
  FROM sessions s
  WINDOW w AS (PARTITION BY s.user_id ORDER BY s.session_start, s.session_id)
)
```

Output: user_session_seq (1 = first observed session) and days_since_prev_session. These feed the return-frequency analysis (Q03b), first-session outcomes (Q23) and the user lifecycle.

Censoring-aware return flags

```
  -- right-censoring: only users observed for at least N days after their first day get a 0/1 value
  CASE WHEN julianday(b.end_date) - julianday(s.first_seen_date) < 7  THEN NULL
       WHEN julianday(nd.next_active_date) - julianday(s.first_seen_date) <= 7  THEN 1 ELSE 0
END,
  CASE WHEN julianday(b.end_date) - julianday(s.first_seen_date) < 14 THEN NULL
       WHEN julianday(nd.next_active_date) - julianday(s.first_seen_date) <= 14 THEN 1 ELSE 0
END,
  CASE WHEN julianday(b.end_date) - julianday(s.first_seen_date) < 28 THEN NULL
       WHEN julianday(nd.next_active_date) - julianday(s.first_seen_date) <= 28 THEN 1 ELSE 0
END,
```

A user first seen 10 days before the end of the data cannot show a 28-day return, so their flag is NULL rather than 0. COUNT(returned_within_28d) then gives the eligible denominator, and AVG() gives the rate over eligible users only.

Cohort grid: recursive CTE

```
WITH RECURSIVE offsets(week_offset) AS (
  SELECT 0
  UNION ALL
  SELECT week_offset + 1 FROM offsets WHERE week_offset < 25
), cohort_size AS (
  SELECT cohort_week, MIN(cohort_week_start) AS cohort_start, COUNT(*) AS cohort_users
  FROM user_summary
  GROUP BY cohort_week
), activity AS (
  SELECT us.cohort_week, w.week_index - us.cohort_week AS week_offset, COUNT(*) AS
active_users
  FROM user_week_activity w
  JOIN user_summary us ON us.user_id = w.user_id
  GROUP BY us.cohort_week, w.week_index - us.cohort_week
)
SELECT cs.cohort_week, cs.cohort_start, o.week_offset, cs.cohort_users,
     COALESCE(a.active_users, 0),
     ROUND(1.0 * COALESCE(a.active_users, 0) / cs.cohort_users, 4)
FROM cohort_size cs
CROSS JOIN offsets o
LEFT JOIN activity a ON a.cohort_week = cs.cohort_week AND a.week_offset = o.week_offset
WHERE cs.cohort_week + o.week_offset <= 26;          -- only weeks that have actually been
observed
```

Output: 351 cohort-week cells. The final WHERE removes cells that have not been observed yet. Without it, recent cohorts would show false zeros in later weeks.

Product metrics: EXISTS lookups and the Wilson lower bound

```
), vb AS (          -- EXISTS uses ix_upp_user_product; joining two DISTINCT subqueries forced a nested scan
  SELECT v.product_id, COUNT(*) AS viewer_buyers
  FROM (SELECT DISTINCT user_id, product_id FROM events_enriched WHERE event_type = 'product_view' AND product_id IS NOT NULL) v
  WHERE EXISTS (SELECT 1 FROM user_product_purchases p WHERE p.user_id = v.user_id AND p.product_id = v.product_id)
  GROUP BY v.product_id

  -- Wilson score lower bound (z = 1.96): ranks conversion without rewarding tiny samples
  CASE WHEN viewers > 0 THEN ROUND(
    ( (1.0 * viewer_buyers / viewers) + 1.9208 / viewers
     - 1.96 * sqrt( (1.0 * viewer_buyers / viewers) * (1 - 1.0 * viewer_buyers / viewers) / viewers + 0.9604 / (1.0 * viewers * viewers) ) )
    / (1 + 3.8416 / viewers), 4) END,
```

The Wilson lower bound gives the lowest view-to-purchase rate consistent, at 95% confidence, with what was observed. Ranking by it stops a product with 3 buyers out of 10 viewers outranking one with 587 out of 4,229.

6.6 Performance tuning

Three steps were initially too slow to use. Each was diagnosed and rewritten:

Step	Problem	Fix	Before → after
DQ11 orders without cart	Correlated NOT EXISTS: the planner chose the event_type index for each order	Pre-aggregated CTE of cart sessions + LEFT JOIN anti-join	150 s → < 1 s
product_performance	Join of two DISTINCT subqueries (viewers × buyers) forced nested scans	EXISTS against user_product_purchases with a composite index	116 s → 0.2 s
category_performance	The same pattern at category level	EXISTS on a (user_id, category_id) index	37 s → 1.3 s

Indexes were designed for the access patterns, for example events (session_id, event_timestamp, event_id) for the per-session window ordering. The full build, including 41 result queries and the exports, runs in about 36 seconds.

6.7 Analytical techniques by query

Query	Technique	Output used for
Q02	Conditional aggregation; median via ROW_NUMBER over the ordered values	Interactions per session
Q03 / Q03b	CASE buckets; window SUM for share and running total; LAG gaps	Return frequency
Q04 / vw_weekly_kpis / vw_daily_kpis	LAG week-over-week %, 4-week and 7-day moving averages (ROWS frames), running totals, LAG(7) same-day-last-week	Engagement trend
Q06 / Q06b	RANK; cumulative share via SUM() OVER (ORDER BY …)	Most-viewed products, concentration

Query	Technique	Output used for
Q07	Closed funnel via multiplied 0/1 flags; LAG step conversion; RANK of drop-offs	User funnel
Q09	UNION ALL of dimensions; percentage-point difference from overall	Where the funnel is weakest
Q10	UNION ALL user timeline; MAX over ROWS BETWEEN UNBOUNDED PRECEDING AND 1 PRECEDING; running SUM	Cart-to-order timing
Q11	LEAD over the session sequence	What users do after add-to-cart
Q13	Average ranks for ties (RANK + (ties − 1)/2); Pearson formula on ranks with sqrt()	Spearman correlation
Q14	Wilson lower bound ranking; minimum sample	Fair conversion ranking
Q15 / Q17b	NTILE quartiles and quintiles; windowed SUM over PARTITION BY category as a benchmark	High-interest, low-conversion products; price effect
Q16 / vw_category_scorecard	DENSE_RANK; window shares	Category purchase activity
Q20	Conditional pivot (MAX(CASE WHEN offset = k …)) over the recursive cohort table	Retention matrix
Q21	Weighted pooling over observed cohorts, three populations	Week 1 / 2 / 4 retention
Q22	LAG change in week-1 retention between cohorts	Cohort quality trend
Q23	UNION ALL of six dimensions; RANK within dimension; pp vs overall	Groups with stronger repeat engagement
Q24	LAG between a buyer's orders; median	Repeat purchase
Q26 / vw_product_scorecard	Catalogue medians via ROW_NUMBER; CASE quadrants; SUM(SUM()) OVER () shares	Popularity vs conversion

6.8 Example: the user funnel (Q07)

```
WITH stages AS (
   SELECT 1 AS stage_no, 'Visited the platform' AS stage, COUNT(*) AS users FROM
user_summary
   UNION ALL SELECT 2, 'Viewed a product',  SUM(ever_viewed_product) FROM
user_summary
   UNION ALL SELECT 3, 'Added to cart',     SUM(ever_viewed_product * ever_added_to_cart)
FROM user_summary
   UNION ALL SELECT 4, 'Began checkout',    SUM(ever_viewed_product * ever_added_to_cart
* ever_began_checkout) FROM user_summary
   UNION ALL SELECT 5, 'Placed an order',   SUM(ever_viewed_product * ever_added_to_cart *
ever_began_checkout * ever_ordered) FROM user_summary
), steps AS (
   SELECT stage_no, stage, users,
          LAG(users) OVER (ORDER BY stage_no)                    AS prev_users
   FROM stages
```

```
)
SELECT stage_no, stage, users,
       ROUND(100.0 * users / prev_users, 2)                                  AS step_conversion_pct,
       prev_users - users                                                    AS users_lost,
       ROUND(100.0 * (prev_users - users) / prev_users, 2)                   AS step_dropoff_pct,
       CASE WHEN stage_no >= 2 THEN ROUND(100.0 * users / (SELECT users FROM stages
WHERE stage_no = 2), 2) END AS pct_of_product_viewers,
       CASE WHEN stage_no > 2 THEN RANK() OVER (ORDER BY CASE WHEN stage_no > 2
THEN 1.0 * (prev_users - users) / prev_users END DESC) END AS dropoff_rank
FROM steps
ORDER BY stage_no;
```

6.9 Example: Spearman correlation in SQL (Q13)

```
WITH r AS (
    SELECT product_id,
           RANK() OVER (ORDER BY views) + (COUNT(*) OVER (PARTITION BY views) - 1) /
2.0                 AS r_views,
           RANK() OVER (ORDER BY units) + (COUNT(*) OVER (PARTITION BY units) - 1) / 2.0
AS r_units,
           RANK() OVER (ORDER BY net_revenue) + (COUNT(*) OVER (PARTITION BY
net_revenue) - 1) / 2.0        AS r_revenue,
           RANK() OVER (ORDER BY view_to_purchase_rate) + (COUNT(*) OVER (PARTITION
BY view_to_purchase_rate) - 1) / 2.0 AS r_conv
    FROM product_performance
), pairs AS (
    SELECT 'views vs units sold' AS pair, r_views AS x, r_units AS y FROM r
    UNION ALL SELECT 'views vs net revenue', r_views, r_revenue FROM r
    UNION ALL SELECT 'views vs view-to-purchase rate', r_views, r_conv FROM r
)
SELECT pair, COUNT(*) AS products,
       ROUND((COUNT(*) * SUM(x * y) - SUM(x) * SUM(y))
           / sqrt((COUNT(*) * SUM(x * x) - SUM(x) * SUM(x)) * (COUNT(*) * SUM(y * y) -
SUM(y) * SUM(y))), 3) AS spearman_rho
FROM pairs
GROUP BY pair;
```

6.10 Views

Section 6 creates 8 views: vw_daily_kpis, vw_weekly_kpis, vw_user_funnel, vw_session_funnel_by_segment, vw_product_scorecard, vw_category_scorecard, vw_cohort_retention and vw_segment_retention. They give the product team stable definitions that a BI tool can query directly. The daily KPI view combines a 7-day moving average, a 7-day conversion rate, running totals and a same-day-last-week comparison:

```
SELECT date, day_name, is_weekend, week_index, promotion_name, active_users, sessions,
product_views,
       cart_sessions, orders, net_revenue,
       ROUND(100.0 * orders / sessions, 2)                                   AS
session_conversion_pct,
       ROUND(AVG(active_users) OVER w7, 1)                                       AS
dau_7d_moving_avg,
       ROUND(100.0 * SUM(orders) OVER w7 / SUM(sessions) OVER w7, 2)                 AS
conversion_7d_pct,
       SUM(orders)      OVER (ORDER BY date ROWS UNBOUNDED PRECEDING)
AS running_orders,
```

```
    SUM(net_revenue) OVER (ORDER BY date ROWS UNBOUNDED PRECEDING)
AS running_net_revenue,
    ROUND(100.0 * (active_users - LAG(active_users, 7) OVER (ORDER BY date))
        / LAG(active_users, 7) OVER (ORDER BY date), 1)                    AS
dau_vs_same_day_last_week_pct
FROM daily
WINDOW w7 AS (ORDER BY date ROWS BETWEEN 6 PRECEDING AND CURRENT
ROW);
```

6.11 Validation queries

Section 7 builds a validation_results table from one CTE of 27 checks, each with an expected and an actual value. Status is PASS when the two agree. Results are in Section 11. The structure:

```
WITH checks (check_id, category, check_name, expected, actual) AS (
  -- record counts
  SELECT 'V01', 'record counts', 'source tables loaded with no row loss (tables failing)', 0,
      (SELECT COUNT(*) FROM load_audit WHERE status <> 'PASS')
  UNION ALL SELECT 'V02', 'record counts', 'events_enriched rows = events rows',
      (SELECT COUNT(*) FROM events), (SELECT COUNT(*) FROM events_enriched)
  UNION ALL SELECT 'V03', 'record counts', 'every event mapped to a calendar week (unmapped
events)', 0,
      (SELECT COUNT(*) FROM events_enriched WHERE week_index IS NULL)
  UNION ALL SELECT 'V04', 'record counts', 'session_summary rows = sessions rows',
      (SELECT COUNT(*) FROM sessions), (SELECT COUNT(*) FROM session_summary)
  UNION ALL SELECT 'V05', 'record counts', 'user_summary rows = users rows',
      (SELECT COUNT(*) FROM users), (SELECT COUNT(*) FROM user_summary)
  -- duplicates and keys
  UNION ALL SELECT 'V06', 'duplicates', 'duplicate event_id in events_enriched', 0,
      (SELECT COUNT(*) - COUNT(DISTINCT event_id) FROM events_enriched)
  UNION ALL SELECT 'V07', 'duplicates', 'duplicate cart additions in cart_add_outcomes', 0,
      (SELECT COUNT(*) - COUNT(DISTINCT event_id) FROM cart_add_outcomes)
  UNION ALL SELECT 'V08', 'join integrity', 'foreign-key violations (PRAGMA
foreign_key_check)', 0,
      (SELECT COUNT(*) FROM pragma_foreign_key_check)
  -- aggregation consistency
  …
```

6.12 Supporting Python (disclosed)

- scripts/run_sql_project.py: executes the SQL and exports results (standard library only).
- scripts/make_charts.py: draws 9 PNG charts from the exported CSVs (pandas, matplotlib). Chart 9 renders genuine sqlite3 shell output.

No figure in this report is computed outside SQL. The only exceptions are two opportunity sizes in Section 9, which are simple products of SQL outputs, with the arithmetic shown.

7. Objective-by-objective analysis

7.1 Objective 1: user engagement

Procedure: headline counts (Q01), per-session interaction statistics (Q02), the distribution of sessions per user and the gaps between sessions (Q03, Q03b), weekly active users with new and returning users and moving averages (Q04, vw_weekly_kpis), engagement by platform and device (Q05), and the most-viewed products (Q06, Q06b).

Measure	Value
Unique users (registered / guest)	30,000 (10,586 / 19,414)
Sessions; sessions per user	89,557; 2.99 (median 1)
Product views (traceable)	311,955 (310,541)
Product interactions per session: mean / median / mean when any	3.99 / 3 / 4.77
Sessions with no product interaction; single-event sessions	16.3%; 22.7%
Average session length	4.43 minutes
Users seen in one session only	55.7%
Median gap between a user's sessions; repeat sessions within 7 days	2.73 days; 78.3%
Weekly active users: range; peak week	1,857–3,441; week of 2025-11-24 (Black Friday)

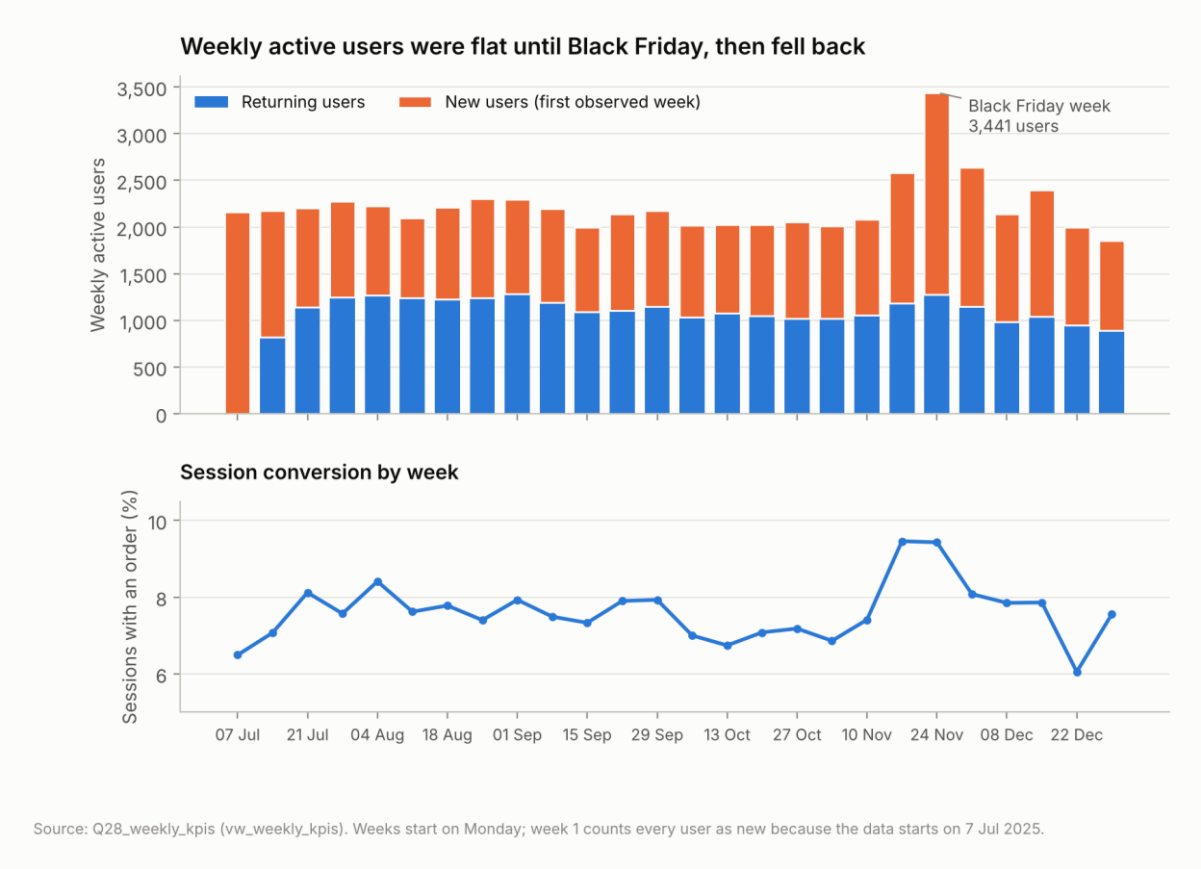


Figure 1. Weekly active users (new vs returning) and session conversion. Source: vw_weekly_kpis.

Interpretation: engagement is steady but shallow for most users. More than half visit once, a fifth of sessions contain a single event, and a typical session has three product interactions. Engagement

is concentrated in a heavy minority: the 6% of users with 10+ sessions include 75.6% who ordered. Product attention is also concentrated: the top 10 products take 22.6% of views. Desktop web sessions are the most engaged and convert at 11.7%, against 5.6% for mobile web, the largest segment.

Rank	Product	Category	Views	% of views	Units rank	View→purchase
1	Voltix Tablet K45	Phones & Tablets	12,399	3.99%	5	6.29%
2	Stride Chinos F33	Fashion	9,337	3.01%	17	2.54%
3	Voltix Soundbar T58	Electronics	8,000	2.58%	9	3.51%
4	Naturelle Vitamin C Serum D49	Health & Beauty	7,527	2.42%	1	13.88%
5	Kitchenix Air Fryer U17	Home & Kitchen	5,910	1.90%	42	1.94%
6	Lagoon Press Novel - Paperback D75	Books & Stationery	5,655	1.82%	23	2.65%
7	Corelink Laptop 14in K13	Computing	5,637	1.82%	34	3.01%
8	Farmfresh Instant Noodles (Carton) X98	Groceries	5,500	1.77%	2	12.76%
9	Stride Ankara Midi Dress G52	Fashion	5,230	1.68%	7	6.03%
10	Voltix Feature Phone V36	Phones & Tablets	5,130	1.65%	87	1.43%

Top 10 most-viewed products (Q06).

7.2 Objective 2: conversion funnel

Procedure: a closed user funnel over the whole window (Q07), in which a user counts at a stage only if they also reached every earlier stage, with purchases taken from orders. A session funnel (Q08) shows behaviour within a single visit. The two are reconciled with event-tracked purchases (Q07b).

Stage	Users	Step conversion	Lost at step	% of product viewers
Visited the platform	30,000	-	-	-
Viewed a product	26,810	89.4%	3,190	100.0%
Added to cart	11,570	43.2%	15,240	43.2%
Began checkout	5,884	50.9%	5,686	21.9%
Placed an order	4,655	79.1%	1,229	17.4%

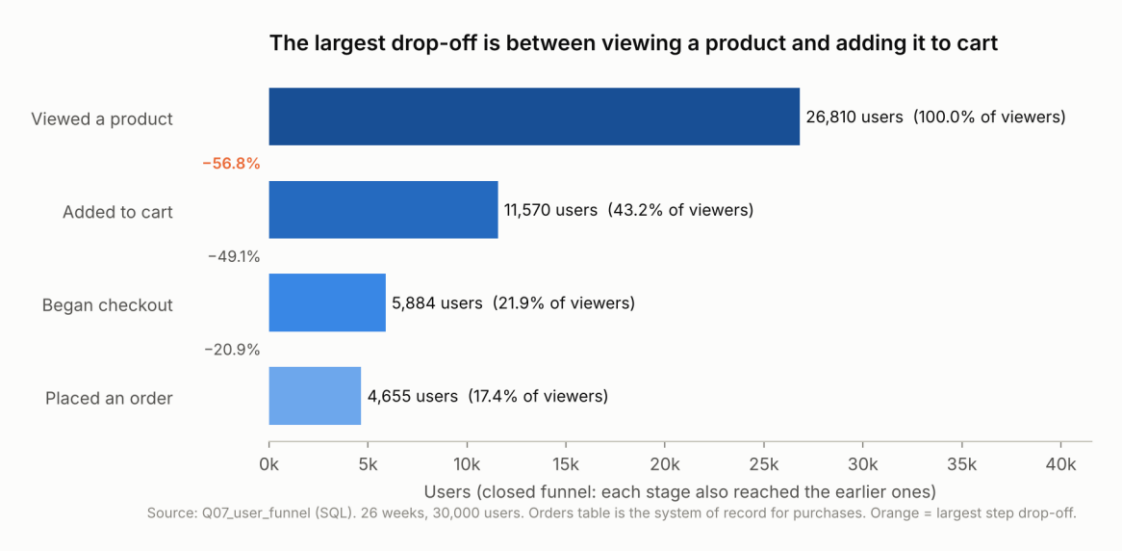


Figure 2. Closed user funnel from product view to order (Q07).

Session stage	Sessions	Step conversion	% of all sessions
All sessions	89,557	–	100.0%
Viewed a product	74,987	83.7%	83.7%
Added to cart	21,361	28.5%	23.9%
Began checkout	6,976	32.7%	7.8%
Placed an order	4,962	71.1%	5.5%

Results: 26,810 users viewed a product, 11,570 added to cart (43.2% of viewers), 5,884 began checkout and 4,655 ordered (17.4% of viewers). Of cart users, 40.2% went on to order. Within a single session, only 4,962 sessions (5.5%) completed the full journey, while 6,896 sessions contained an order. That difference is the cross-session journeys.

Interpretation: the funnel loses most users at its first step. Per session the picture looks worse (only 28.5% of viewing sessions add to cart), because users spread a purchase across visits. The user funnel is the right lens for the Product Manager's question.

7.3 Objective 3: abandonment

Procedure: the largest step drop-off is ranked in Q07. Segment funnels (Q09) locate where each step is weakest. LEAD shows the event that follows an add-to-cart (Q11). Q10 measures how long carts take to become orders. Q12 measures checkout abandonment by platform and device, and Q12b measures overall cart abandonment with a censoring-aware 7-day window.

Drop-off point	Measure	Value
View → cart	product viewers who never add to cart	15,240 users (56.8%): largest
Cart → checkout	cart users who never begin checkout	5,686 users (49.1%)
Checkout → order	checkout users who never order	1,229 users (20.9%)
Checkout sessions	end without an order	2,920 of 9,816 (29.7%)
Cart additions	not bought within 7 days	67.9% of 28,001 with an outcome

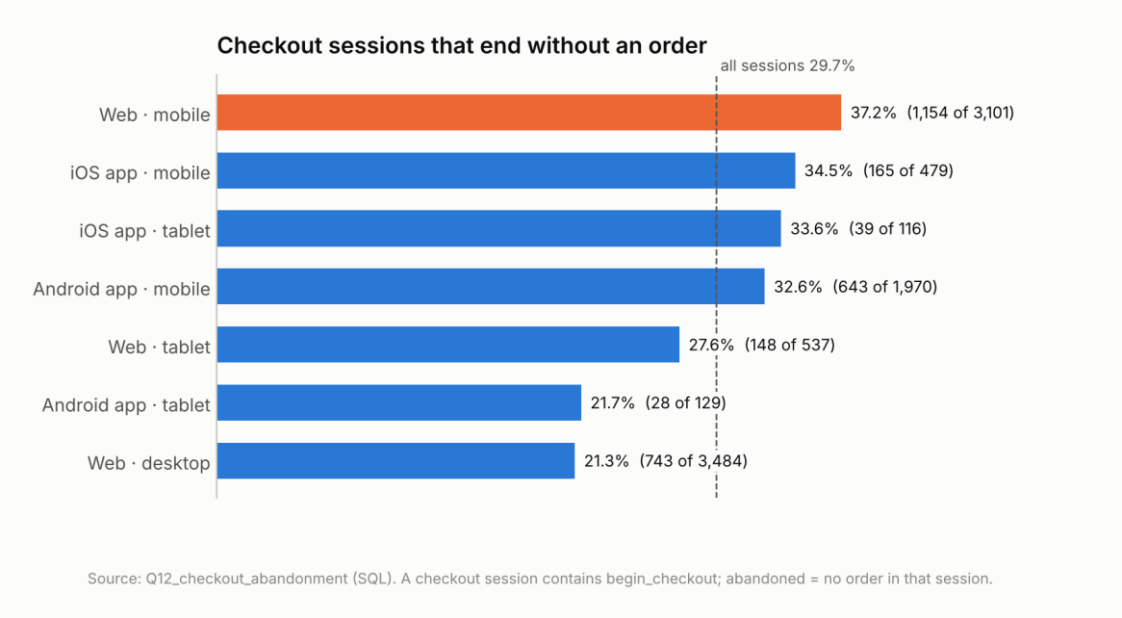


Figure 3. Checkout abandonment by platform and device (Q12).

Next event after add-to-cart	Events	%
product_view	11,182	38.6%
page_view	6,310	21.8%
search	4,931	17.0%
(session ended)	3,313	11.4%
begin_checkout	1,901	6.6%
remove_from_cart	1,304	4.5%

Time from last add-to-cart to order	Orders	% of orders	Cumulative
Same session	4,960	71.9%	71.9%
Earlier session, < 1 day	379	5.5%	77.4%
1-3 days	384	5.6%	83.0%
3-7 days	518	7.5%	90.5%
More than 7 days	643	9.3%	99.8%
No earlier add-to-cart	12	0.2%	100.0%

Interpretation: abandonment has three distinct shapes:

- **Browsing without commitment**, at the top of the funnel. Paid social is the extreme case: 18.0% view-to-cart against 34.3% for email.
- **Shortlisting.** After an add-to-cart, 38.6% of users view another product next and only 6.6% go to checkout. 71.9% of orders follow the last add-to-cart in the same session, and 9.3% come more than 7 days later.
- **Friction at checkout**, concentrated on mobile web: 37.2% abandoned against 21.3% on desktop web.

7.4 Objective 4: product and category performance

Procedure: product rankings by views and units (Q06, Q13b); Spearman rank correlations between popularity and outcomes (Q13); popularity/conversion quadrants around the catalogue medians

(Q26); conversion ranking by Wilson lower bound with at least 500 viewers (Q14); high-interest, low-conversion products by NTILE quartiles against a category benchmark (Q15); category activity (Q16); and product and category cart abandonment with at least 100 observed additions (Q17, Q17b).

Rank correlation (250 products)	Spearman ρ
views vs net revenue	0.599
views vs units sold	0.517
views vs view-to-purchase rate	-0.177

Quadrant	Products	% of views	% of units	Pooled view→purchase
Traffic without sales: high traffic, low conversion	75	47.4%	20.6%	2.51%
Star: high traffic, high conversion	51	33.7%	53.4%	7.42%
Hidden gem: low traffic, high conversion	75	11.4%	21.8%	7.04%
Low priority: low traffic, low conversion	49	7.5%	4.2%	2.44%

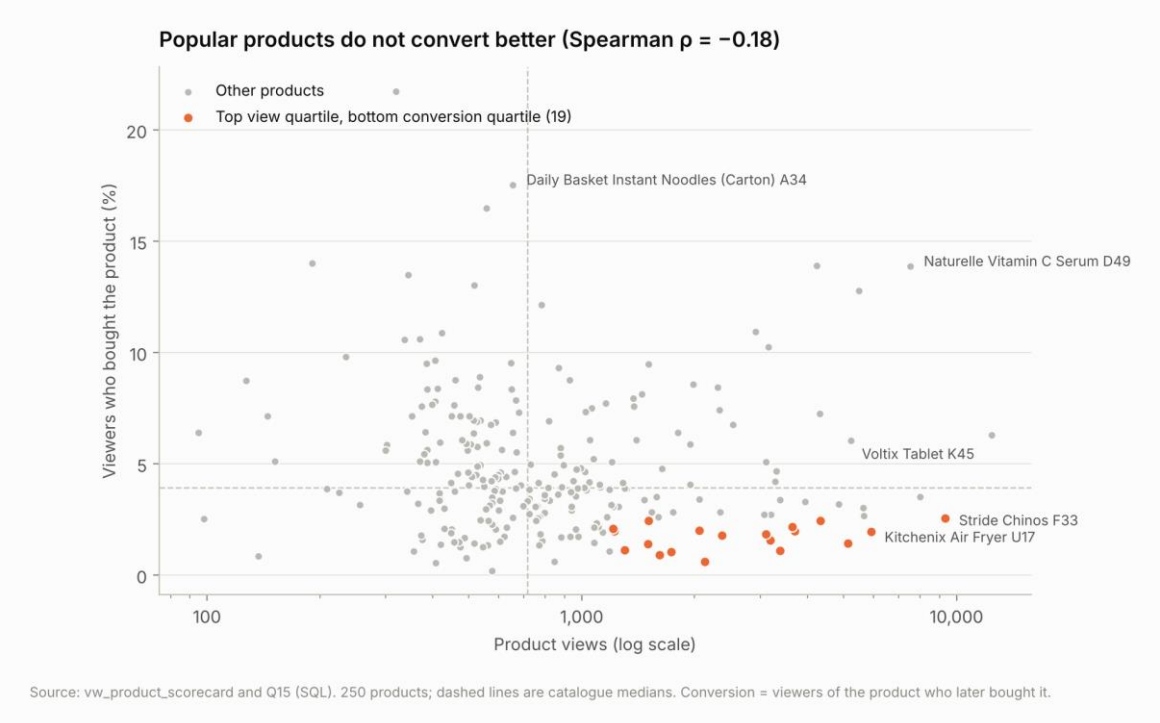


Figure 4. Product views vs view-to-purchase conversion; orange = top view quartile and bottom conversion quartile (Q15).

Do popular products necessarily generate the most purchases? No. Popular products sell more units in absolute terms (ρ = 0.52), but popularity and conversion are weakly negatively related (ρ = -0.18). Only 8 of the 20 most-viewed products are among the 20 best sellers. The "traffic without sales" quadrant takes 47.4% of views but delivers 20.6% of units.

Product (≥ 500 viewers)	Category	Viewers	Conversion	Wilson lower
Daily Basket Instant Noodles (Carton) A34	Groceries	553	17.54%	14.60%
Naturelle Vitamin C Serum D49	Health & Beauty	4,229	13.88%	12.87%
Glowa Shea Body Butter V48	Health & Beauty	2,696	13.91%	12.65%
Farmfresh Instant Noodles (Carton) X98	Groceries	3,283	12.76%	11.66%
Playmore Baby Wipes 3-Pack F67	Baby, Kids & Toys	643	12.13%	9.83%
Ori Botanics Shea Body Butter U44	Health & Beauty	2,102	10.94%	9.68%
Daily Basket Tea Bags 50ct V27	Groceries	2,136	10.25%	9.04%
Mama's Pantry Parboiled Rice 5kg K20	Groceries	1,183	9.47%	7.93%

Highest view-to-purchase conversion, ranked by the Wilson lower bound (Q14, top 8).

High views, low conversion	Views	Conversion	Category rate	Cart abandon.	Buyer gap
Stride Chinos F33	9,337	2.54%	4.14%	86.6%	83
Melanin Lab Shea Body Butter X82	1,302	1.13%	8.23%	77.8%	75
Voltix Feature Phone V36	5,130	1.43%	3.35%	61.1%	65
Bumble Kids Sneakers P52	3,642	2.17%	5.31%	72.9%	61
Kente & Co Handbag N33	4,331	2.43%	4.14%	69.1%	51
Kelvo Chest Freezer 200L D32	3,377	1.11%	2.76%	90.3%	39
Axiom Feature Phone Y80	3,701	1.98%	3.35%	71.5%	36
Kitchenix Air Fryer U17	5,910	1.94%	2.76%	87.4%	29

Q15, top 8 of 19. The category rate is the pooled product-level rate of the product's category (a windowed SUM). Buyer gap = viewers × (category rate − product rate); the 19 products sum to 656. This sizes the gap; it is not a forecast.

Category	Orders	Buyers	Units	Conversion	% of orders	% of revenue
Health & Beauty	1,699	1,409	3,552	15.5%	18.5%	3.6%
Groceries	1,505	1,262	3,435	15.9%	16.4%	3.4%
Fashion	1,414	1,207	2,131	9.8%	15.4%	6.1%
Phones & Tablets	994	904	1,088	8.0%	10.8%	35.4%
Electronics	828	750	1,165	7.7%	9.0%	7.4%
Baby, Kids & Toys	773	694	1,130	10.3%	8.4%	3.9%
Home & Kitchen	592	546	819	6.5%	6.5%	3.1%
Books & Stationery	580	517	1,118	8.6%	6.3%	0.8%
Sports & Fitness	408	378	557	7.8%	4.5%	2.5%
Computing	370	347	391	5.2%	4.0%	33.8%

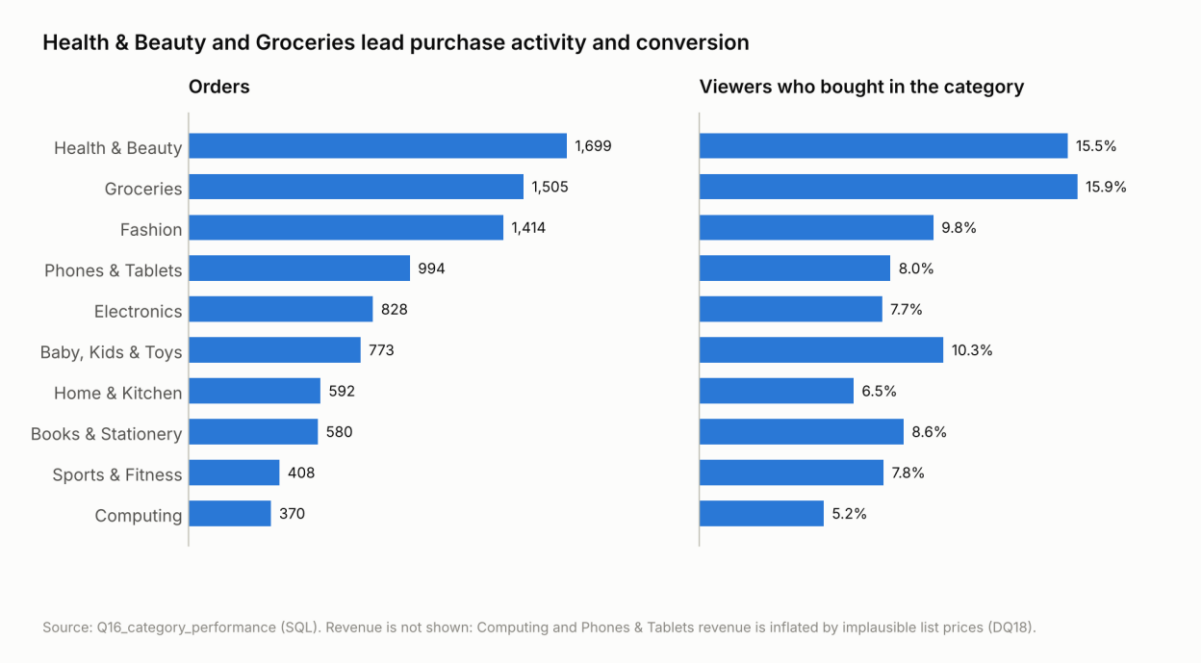


Figure 5. Orders and viewer conversion by category (Q16).

Which categories have the highest purchase activity? Health & Beauty (1,699 orders) and Groceries (1,505), which also have the highest viewer conversion (15.5% and 15.9%). Phones & Tablets and Computing lead net revenue (35.4% and 33.8%), but on few units and with implausible prices (DQ18). Revenue rankings are therefore indicative.

Highest cart abandonment (≥ 100 observed additions)	Category	Observed	Abandoned	vs category
Lekki Thread Men's Kaftan Set K73	Fashion	137	93.4%	+20.1 pp
Kente & Co Sneakers K43	Fashion	135	91.1%	+17.8 pp
Playmore Baby Stroller X62	Baby, Kids & Toys	108	90.7%	+25.1 pp
Kelvo Chest Freezer 200L D32	Home & Kitchen	207	90.3%	+15.9 pp
Kitchenix Air Fryer U17	Home & Kitchen	428	87.4%	+12.9 pp
Stride Chinos F33	Fashion	881	86.6%	+13.3 pp
Zentel Smartphone P32	Phones & Tablets	101	81.2%	+13.8 pp
Sonara Inverter 1.5kVA J13	Electronics	421	81.0%	+8.2 pp

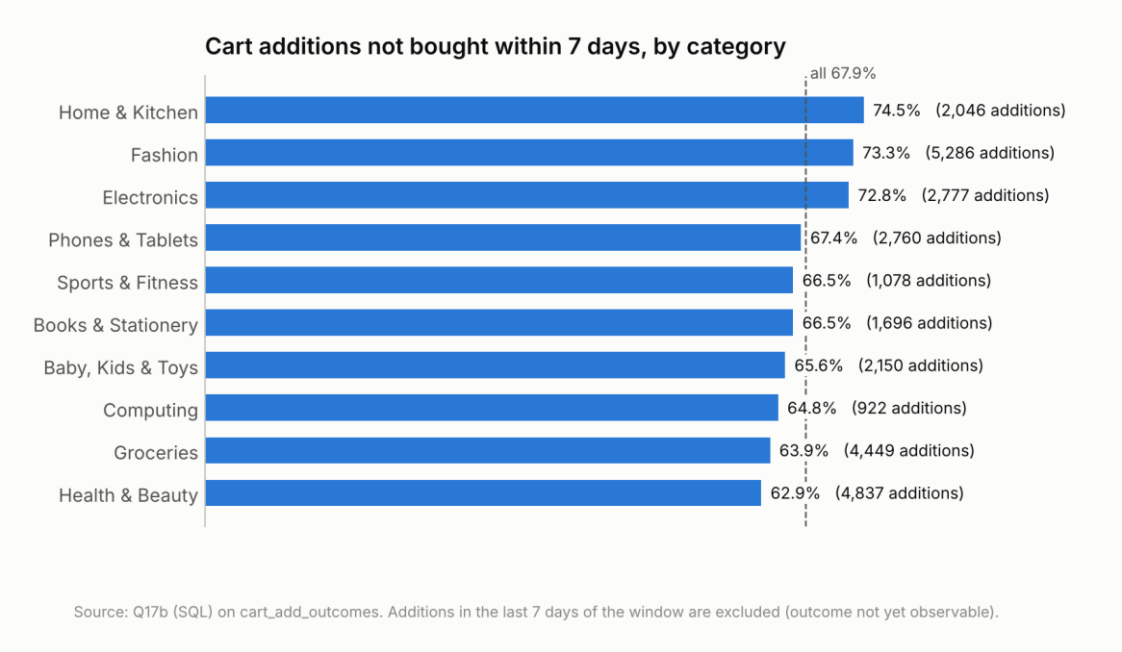


Figure 6. Cart additions not bought within 7 days, by category (Q17b).

7.5 Objective 5: returning users and retention

Procedure: return on a later day and within 7 / 14 / 28 days among eligible users (Q19); a weekly cohort retention matrix built from the recursive cohort table (Q20); pooled week-1, week-2 and week-4 retention for three populations, to handle left-censoring (Q21); the cohort quality trend (Q22); return and repeat buying by user group (Q23); and repeat purchase (Q24).

Measure	Value
Users who returned on a later day	12,667 (42.2%), median 4 days to return
Returned within 7 / 14 / 28 days (eligible users)	31.4% / 39.4% / 43.2%
Week 1 / 2 / 4 retention: all users	26.5% / 19.3% / 11.0%
Week 1 / 2 / 4: cohorts 3–26	25.1% / 17.8% / 9.7%
Week 1 / 2 / 4: users not registered before the window	20.0% / 13.8% / 6.8%
Repeat buyers; share of orders; median days between orders	28.4% of buyers; 51.7%; 10.1 days

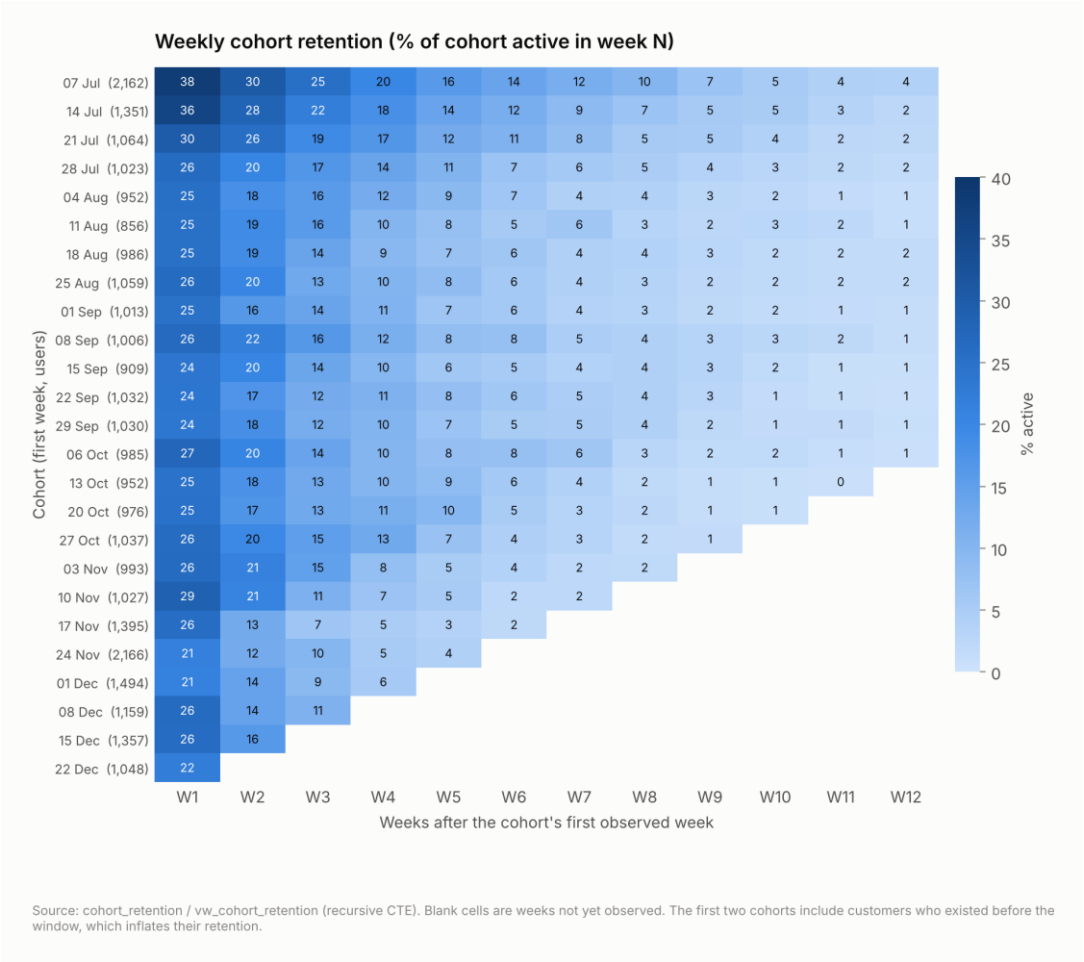


Figure 7. Weekly cohort retention, weeks 1–12 (cohort_retention).

Group	Users eligible (28 d)	Return within 28 days	vs overall	Repeat buyers
Channel: email	1,222	71.7%	+28.5 pp	39.2%
Channel: direct	3,452	61.8%	+18.6 pp	34.2%
Channel: organic_search	6,748	46.6%	+3.4 pp	27.1%
Channel: paid_search	3,455	40.3%	-2.9 pp	22.8%
Channel: affiliate	2,225	34.8%	-8.4 pp	26.4%
Channel: paid_social	5,419	25.0%	-18.2 pp	21.5%
loyalty member: member	1,833	88.5%	+45.3 pp	45.2%
loyalty member: not a member	23,635	39.7%	-3.5 pp	23.5%
registration: registered	9,388	74.2%	+31.0 pp	28.4%
registration: guest	16,080	25.1%	-18.1 pp	-
primary device: desktop	6,265	45.5%	+2.3 pp	33.6%
primary device: mobile	17,413	42.5%	-0.7 pp	25.4%

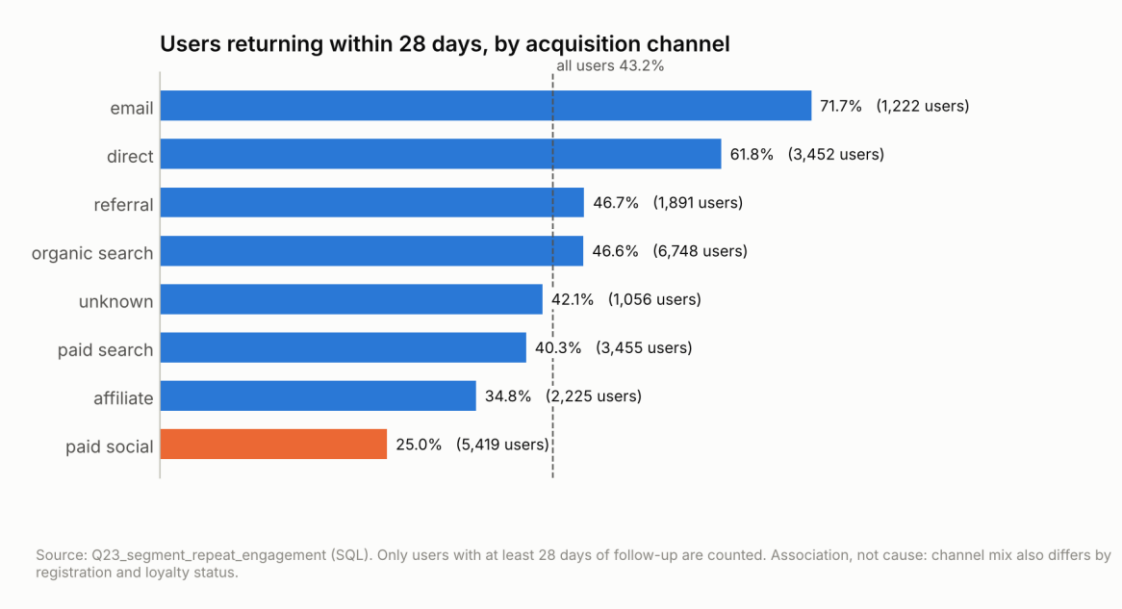


Figure 8. 28-day return by acquisition channel (Q23).

Interpretation: retention is decided early. The median user who returns does so within 4 days, and retention roughly halves between week 1 and week 4. Week-1 retention was stable at about 24-29% for the cohorts from late July to mid-November. The Black Friday cohorts (1,395 and 2,166 users) had lower week-4 retention (4.9%, 5.3%). This fits deal-driven acquisition, but those cohorts reach week 4 during the late-December slowdown, so the two cannot be separated. Email-acquired, registered and loyalty users return far more. Registration and loyalty are recorded at extract date, however, so the direction of these associations is unknown.

7.6 Objective 6: opportunities

Procedure: Q25 consolidates the size of each leak. Promotion periods are compared per day (Q18). Two gaps are sized by applying a benchmark rate to a segment (shown below). These sizes indicate scale and are not forecasts.

Priority	Opportunity	Size	Base
1	Viewers who never add to cart	15,240	56.8% of product viewers
2	Cart users who never reach checkout	5,686	49.1% of cart users
3	Checkout sessions that end without an order	2,920	29.7% of checkout sessions
4	Cart additions not bought within 7 days	19,020	67.9% of cart additions with an outcome
5	High-traffic, low-converting products	19	7.6% of the catalogue (top view quartile, bottom conversion quartile)
6	Users seen in only one session	16,704	55.7% of all users

Mobile-web checkout gap: 3,101 checkout sessions × (37.21% - 21.33%) ≈ **493 orders** over 26 weeks, 7.1% of all orders, if mobile web completed checkouts at the desktop-web rate.

High-traffic, low-conversion products: the 19 products in Q15 would add about **656 product purchases** (buyer-product pairs) if they converted at their category rate.

Period	Days	Sessions per day	View→cart	Session conversion	Avg net order value
No promotion	122	450	28.1%	7.44%	82,554
Back-to-School Sale	21	506	28.0%	7.61%	95,488
Independence Day Deals	7	476	28.8%	7.93%	88,618
Black Friday	10	864	32.0%	9.36%	78,570
Cyber Week Tech Deals	5	678	27.4%	7.73%	83,034
Festive Season Sale	13	531	29.1%	7.59%	68,357
New Year Clearance	4	437	26.7%	8.13%	55,247

7.7 Objective completion

Objective	Status	Main evidence
1 Engagement	Complete	Q01–Q06b, Q27, Q28; Figure 1
2 Conversion funnel	Complete	Q07, Q07b, Q08; Figure 2
3 Abandonment stages	Complete	Q07 (ranked drop-offs), Q09–Q12b; Figure 3
4 Product and category performance	Complete. Revenue ranking is indicative only (DQ18)	Q13–Q17b, Q26; Figures 4–6
5 Returning users and retention	Complete, with censoring corrections	Q19–Q24; Figures 7–8
6 Opportunities	Complete. Sizes are not forecasts	Q25, Q18, Sections 8–9

8. Key findings

1. **The top of the funnel is the biggest leak.** 56.8% of product viewers (15,240 users) never add to cart. Cart → checkout loses 49.1%, and checkout → order loses 20.9%.
2. **Mobile-web checkout loses intent.** 37.2% of mobile-web checkout sessions end without an order, against 21.3% on desktop web. Mobile web is the largest segment and converts at 5.6%.
3. **The cart is a shortlist and journeys span sessions.** 38.6% of add-to-carts are followed by another product view. 67.9% of additions are not bought within 7 days, and 28% of orders complete a journey from an earlier session.
4. **Popularity does not mean conversion.** The rank correlation of views with conversion is -0.18. 75 products take 47.4% of views but 20.6% of units, while 75 "hidden gems" convert at 7.04% on 11.4% of views.
5. **Consumables drive purchase activity.** Health & Beauty and Groceries lead orders and conversion, and have the lowest cart abandonment.
6. **Retention is decided in the first week.** Week-1 retention is 26.5% and week-4 is 11.0%. The median return comes after 4 days.
7. **Channel quality differs.** 28-day return is 71.7% for email-acquired users and 25.0% for paid social. Paid-social sessions convert at 3.5%.
8. **Black Friday** doubled daily sessions (864 vs 450) and raised conversion (9.4% vs 7.4%), but its cohorts retained worse. That comparison is confounded with the December slowdown.
9. **Repeat buyers matter.** 28.4% of buyers placed 51.7% of orders.
10. **The data itself needs fixing.** The event stream misses 2.99% of orders, and 12 products carry implausible prices.

9. Business recommendations

Each recommendation follows from a finding above. Associations are treated as hypotheses to test, not as proven effects.

Horizon	Recommendation	Evidence	Success measure
Immediate (0-4 weeks)	Fix purchase-event tracking; schedule the events-vs-orders reconciliation (V11, V26)	DQ08, Q07b	0 orders missing from events
Immediate	Audit the 12 implausible catalogue prices	DQ18, Q16	DQ18 = 0
Immediate	Audit the mobile-web checkout (form length, payment options, speed, errors) and fix obvious friction	Q12, Q09	Mobile-web checkout abandonment
Immediate	Review the 19 high-traffic, low-conversion product pages (price, images, size/spec, reviews, stock)	Q15, Q26	Product view-to-purchase
Medium (1–3 months)	A/B test the mobile-web checkout redesign	Q12	Checkout → order on mobile web
Medium	A/B test a persistent cross-device cart, save-for-later, and a cart reminder 24–72 h after an addition	Q10, Q11, Q12b	7-day cart conversion
Medium	First-week re-engagement for new users, triggered by inactivity	Q19, Q21	Week-1 and week-4 retention of new cohorts
Medium	Judge paid social on downstream add-to-cart, orders and 28-day return; review targeting	Q09, Q23	Orders and returns per paid-social user
Medium	Give more placement to hidden-gem products; test replenishment reminders on consumables	Q26, Q14, Q16	Units per view; repeat-buyer rate
Long term (3-12 months)	Randomised tests of registration and loyalty prompts before scaling them	Q23	Incremental retention vs control
Long term	Track promotion cohorts after each event; a cohort dashboard on vw_cohort_retention and vw_weekly_kpis	Q22, Q18	Adopted by the PM team
Long term	Cross-device identity, event data contracts, and marketing cost data for channel return	DQ11, Q23	Stitched journeys; cost per retained buyer

10. Technical and analytical limitations

10.1 Observed data limitations

- **The data is synthetic**: the patterns demonstrate the method only.
- Tracking loss affects 3% of orders, so purchases are counted from orders.
- 1,414 product views and 127 cart additions are untraceable to a product.
- 12 products have implausible prices, which makes revenue rankings indicative.
- Registration, loyalty and channel are recorded at extract date.
- The currency is not stated, and there is no cost, margin or spend data.

10.2 Methodological limitations

- The analysis is observational: segment differences are associations. Registration and loyalty effects may be reverse causation.
- Left-censoring inflates cohorts 1-2, so retention is also reported excluding them. Right-censoring is handled with NULL flags and observed-only cohort cells.
- The 7-day abandonment window counts the 9.3% of orders that come later than 7 days as abandoned.
- Viewer-buyer attribution means a viewer ordered the same product at any time in the window. It does not show that the view caused the order.
- Black Friday cohort retention is confounded with December seasonality.
- Minimum samples (500 viewers, 100 cart additions) leave smaller products unranked.
- Opportunity sizes are scale indicators, not forecasts.

10.3 Contextual limitations

- 26 weeks cannot show year-over-year seasonality.
- There are no page-level, search-term, error or checkout-step events, so the analysis shows where users drop off but not why.
- Users who switch devices without logging in appear as separate IDs.

10.4 Assumptions

- Orders are the authoritative record of purchases. A net order is not cancelled or returned; revenue = items_subtotal, excluding shipping.
- A returning user is active on a later calendar day. Weeks run Monday to Sunday (calendar table). Prices are in NGN.
- A NULL signup_date means a guest. A signup before 7 Jul 2025 means a known existing customer. No bot traffic is present.

10.5 Information that would strengthen the analysis

The most useful additions would be page-level and checkout-step events, A/B test assignments, marketing spend by channel, product cost and margin, a reference price list, and a longer history.

11. Validation and quality assurance

11.1 Reconciliation and business-rule checks

ID	Category	Check	Expected	Actual	Status
V01	record counts	source tables loaded with no row loss (tables failing)	0	0	PASS
V02	record counts	events_enriched rows = events rows	590,648	590,648	PASS
V03	record counts	every event mapped to a calendar week (unmapped events)	0	0	PASS
V04	record counts	session_summary rows = sessions rows	89,557	89,557	PASS
V05	record counts	user_summary rows = users rows	30,000	30,000	PASS
V06	duplicates	duplicate event_id in events_enriched	0	0	PASS
V07	duplicates	duplicate cart additions in cart_add_outcomes	0	0	PASS
V08	join integrity	foreign-key violations (PRAGMA foreign_key_check)	0	0	PASS
V09	aggregation	sum of session event counts = events	590,648	590,648	PASS
V10	aggregation	sum of user session counts = sessions	89,557	89,557	PASS
V11	aggregation	orders in session_summary = orders	6,896	6,896	PASS
V12	aggregation	product views in product_performance = traceable product_view events	310,541	310,541	PASS
V13	aggregation	units in product_performance = units in order_items	15,386	15,386	PASS
V14	aggregation	weekly cohort sizes sum to all users	30,000	30,000	PASS
V15	reconciliation	net revenue: user_summary = orders table	512,197,600	512,197,600	PASS
V16	reconciliation	net revenue: product_performance = order_items of net orders	512,197,600	512,197,600	PASS
V17	reconciliation	net revenue: category_performance = product_performance	512,197,600	512,197,600	PASS
V18	reconciliation	closed user funnel final stage = all buyers (every buyer passed every stage)	4,655	4,655	PASS
V19	reconciliation	weekly orders in vw_weekly_kpis = orders	6,896	6,896	PASS
V20	reconciliation	daily net revenue in vw_daily_kpis = user_summary net revenue	512,197,600	512,197,600	PASS
V21	business rule	user funnel never increases from one stage to the next (violations)	0	0	PASS
V22	business rule	week-0 retention is 100% for every cohort (violations)	0	0	PASS

ID	Category	Check	Expected	Actual	Status
V23	business rule	retention and conversion rates within [0, 1] (violations)	0	0	PASS
V24	business rule	every order session contains a begin_checkout event (orders without)	0	0	PASS
V25	business rule	a same-session purchase is also a purchase within 7 days (violations)	0	0	PASS
V26	business rule	event-tracked buyers do not exceed order-based buyers (excess)	0	0	PASS
V27	business rule	censored return flags only for users with too little follow-up (violations)	0	0	PASS

11.2 Further QA performed

QA step	How it was tested	Result
Script runs end to end in the sqlite3 shell	sqlite3 -bail with the full file on a fresh database	Exit code 0, 33.4 s
Script runs through the runner	run_sql_project.py --fresh	70 blocks, 35.5 s, no errors
Two execution paths agree	MD5 of the full contents of every table and view in both databases	30 objects compared, 0 differ
Checks can detect errors	On a copy: deleted one ordered session and added 1 unit to one product, then re-ran Section 7	V04, V09, V11, V13, V19 and V20 failed, as expected
Raw data unchanged	MD5 of the raw file before and after	451c337ac01d6286b2a1575b246ed178 = 451c337ac01d6286b2a1575b246ed178
Load completeness	Source vs loaded row counts	10/10 PASS
Constraint integrity	PK / FK / UNIQUE / CHECK at load; PRAGMA foreign_key_check	No rejections; 0 violations
Numbers in documents	The report is generated from the exported CSVs; README and docs figures cross-checked against the same files	No hand-typed results in this report
Charts	Each chart rendered and inspected for overlap, labels and correctness	9 charts from actual outputs

11.3 Issues that remain unresolved

- The implausible prices (DQ18) cannot be corrected without a reference price list, so revenue rankings remain indicative.
- Seasonality and acquisition quality cannot be separated for the Black Friday cohorts.
- The causal effect of registration, loyalty and channel is unknown without experiments.

12. Lessons and professional competencies demonstrated

Competency	Where it shows
Business problem-solving	Translated the Product Manager's questions into precise metric definitions and an objective-to-query map. Ended with prioritised, sized opportunities
Data preparation	Constrained schema, staging with type conversion, a 19-check DQ register, keep-and-flag cleaning, derived analysis layer
Tool proficiency (SQL)	Multi-CTE pipelines, recursive CTE, ROW_NUMBER, RANK, DENSE_RANK, NTILE, LAG, LEAD, running totals, moving averages, custom frames, FILTER, EXISTS, views, PRAGMAs, ATTACH
Analytical thinking	Chose orders over events, and the user funnel over the session funnel, on evidence. Separated three kinds of abandonment
Statistical reasoning	Medians, Spearman correlation and Wilson bounds in SQL. Censoring corrections. Minimum samples. Caution about confounding and reverse causation
Data modelling	Grain-aware derived tables (event, session, user, user-week, cohort, cart addition, product, category) with keys and indexes
Performance engineering	Diagnosed planner problems; rewrote queries as anti-joins and indexed EXISTS (116 s → 0.2 s)
Quality assurance	27 automated checks, a seeded-error test, two-path equivalence, raw-file checksum
Reporting and communication	Finding comments in the SQL, charts, this report, README, recommendations by horizon

Lessons

- Reconcile the event stream with the transactional system first. It decided how every purchase metric was defined.
- In SQLite, qualify DROP statements when another database is attached, and attach sources read-only.
- Censoring is easy to miss in retention. NULL-able outcome flags make it explicit and keep denominators honest.
- Read the query plan before adding hardware: the slowest steps were planner choices, not data volume.

13. Conclusion

The project turned 590,648 raw events into a validated SQL analytics layer and answered every question in the brief.

- **Engagement** is steady but shallow: 55.7% of users visit once.
- **Abandonment** is greatest at the first funnel step: 56.8% of viewers never add to cart.
- **Checkout** is weakest on mobile web: 37.2% abandoned, against 21.3% on desktop.
- **Popular products** do not convert better, and consumables lead purchase activity.
- **Retention** is decided in the first week: 26.5% at week 1 and 11.0% at week 4.

For the Product Manager this gives a clear order of work: fix tracking and catalogue data, remove mobile-web checkout friction, repair or re-merchandise high-traffic products that do not convert, make carts persistent, and win users back in their first week. Each change should be tested, because the data shows associations, not causes. The SQL file runs end to end in about 36 seconds, and its 27 validation checks all pass, so the analysis can be refreshed and trusted as new data arrives.

Appendix A. Result files

File (outputs/results/)	Question or content	Rows
Q00_load_audit.csv	Source rows vs loaded rows for every table (load audit)	10
Q00_dq_register.csv	The data-quality register, DQ01–DQ19, with affected rows and actions	19
Q00_column_completeness.csv	Column-level NULL profile of the event table (completeness by event type).	7
Q00_outlier_profile.csv	Distribution of sessions per user and events per session: percentiles from ROW_NUMBER (SQLite has no PERCENTILE_CONT).	2
Q00_derived_table_sizes.csv	Row counts of the derived tables built in Section 4	9
Q01_platform_overview.csv	How many unique users interacted with the platform, and how many product views were recorded?	1
Q02_session_engagement.csv	What is the average number of product interactions per session?	1
Q03_return_frequency.csv	How frequently do users return? Distribution of users by number of sessions, with running cumulative share.	6
Q03b_time_between_sessions.csv	When users come back, how soon? Gaps between consecutive sessions (LAG, already stored as days_since_prev_session).	1
Q04_weekly_engagement_trend.csv	How does engagement move week by week? WAU, new vs returning users, week-over-week change (LAG), 4-week moving average (window AVG) and the	26
Q05_engagement_by_platform.csv	Does engagement differ by platform and device? Share of sessions, interactions and conversion per segment, compared with the overall rate (w	7
Q06_top_viewed_products.csv	Which products receive the most views? Top 15 with share of all traceable views and running cumulative share.	15
Q06b_view_concentration.csv	How concentrated is attention? Share of views captured by the top 10, 25 and 50 products (Pareto check).	3
Q07_user_funnel.csv	How many users viewed products, added to cart, began checkout and purchased? Where is the largest drop? Closed funnel: each stage counts use	5
Q07b_purchase_reconciliation.csv	Reconciles the order-based purchase stage with the event stream (tracking-loss check from DQ08).	1
Q08_session_funnel.csv	Session-level funnel (all stages within the same session). Shows how much of the drop-off is "not yet": 28% of orders happen in a session wi	5

File (outputs/results/)	Question or content	Rows
Q09_funnel_by_segment.csv	Where does the drop-off differ? Session step rates by platform/device and by traffic source, with the difference from the overall rate in pe	14
Q10_cart_to_order_timing.csv	How long do purchase journeys take? For each order, the time since the user's most recent add-to-cart.	6
Q11_after_add_to_cart.csv	What do users do immediately after adding to cart? Next event in the same session (LEAD).	6
Q12_checkout_abandonment.csv	Of sessions that reach checkout, how many end without an order, and does it differ by platform and device?	8
Q12b_cart_abandonment_overall.csv	Cart abandonment at the level of individual cart additions (traceable products, 7-day outcome window).	1
Q13_popularity_vs_purchases.csv	Do popular products necessarily generate the most purchases? Spearman rank correlation between views and units sold (and between views and c	3
Q13b_top_viewed_vs_top_sellers.csv	The overlap between the 20 most viewed and the 20 best-selling products (by units).	1
Q14_top_conversion_products.csv	Which products have the highest view-to-purchase conversion? Products with at least 500 viewers, ranked by the 95% Wilson lower bound so tha	15
Q15_high_interest_low_conversion.csv	Opportunity list: products in the top quartile for views but the bottom quartile for conversion (NTILE). Benchmark = the pooled product-leve	19
Q16_category_performance.csv	Which categories have the highest purchase activity? Orders, buyers, units and net revenue with ranks.	10
Q17_cart_abandonment_products.csv	Which products experience relatively high cart abandonment? Products with at least 100 cart additions that have a 7-day outcome, compared wi	15
Q17b_abandonment_by_category_and_price.csv	Is abandonment linked to price? Cart abandonment by category and by list-price band (NTILE price quintiles).	15
Q18_promotion_periods.csv	How do sessions behave during promotion periods? Descriptive comparison of promotion days with other days (not a causal estimate: promotion	7
Q19_returning_users.csv	How many users returned after their first observed interaction? Return within 7/14/28 days uses only users with enough follow-up (right-cens	1
Q20_cohort_retention_matrix.csv	How does activity change across weekly cohorts? Share of each first-seen weekly cohort active in each later week (pivot by conditional aggre	26
Q21_retention_week_1_2_4.csv	What proportion of users return after one, two or four weeks? Week-N retention = active in calendar week (cohort week + N), pooled over ever	9

File (outputs/results/)	Question or content	Rows
Q22_cohort_quality_trend.csv	Are later cohorts as engaged as earlier ones? Per cohort: size, week-1 and week-4 retention, orders per user in the first 4 weeks, and the c	26
Q23_segment_repeat_engagement.csv	Which user groups show stronger repeat engagement? For each grouping: users, 28-day return rate (users with 28 days of follow-up), average a	31
Q24_repeat_purchase.csv	Repeat buying: buyers with 2+ orders and the time between consecutive orders (LAG over each buyer's orders).	1
Q25_opportunity_summary.csv	Consolidated opportunity sizes from the funnel, abandonment, product and retention outputs	6
Q26_product_quadrants.csv	Do popular products necessarily generate the most purchases? Products and views by quadrant.	4
Q27_daily_kpis.csv	Daily KPIs from vw_daily_kpis (7-day moving averages, running totals, same day last week)	182
Q28_weekly_kpis.csv	Weekly KPIs from vw_weekly_kpis (new vs returning, week-over-week change, 4-week moving average)	26
Q99_validation_results.csv	All 27 validation checks with expected value, actual value and status	27
Q99_validation_summary.csv	Number of validation checks passed and failed	1

Appendix B. Repository structure

```
ecommerce-product-analytics-sql/
├── data/
│   ├── processed/
│   │   ├── cart_add_outcomes.csv
│   │   ├── category_scorecard.csv
│   │   ├── cohort_retention.csv
│   │   ├── ecommerce_analytics.db   (build artifact, git-ignored)
│   │   ├── events_enriched.csv.gz
│   │   ├── product_scorecard.csv
│   │   ├── session_funnel_by_segment.csv
│   │   ├── session_summary.csv
│   │   └── user_summary.csv
│   ├── raw/
│   │   └── ecommerce_clean.sqlite
│   └── data_dictionary.md
├── docs/
│   ├── data_quality_and_cleaning_log.md
│   ├── insights_and_recommendations.md
│   ├── kpi_definitions.md
│   ├── limitations.md
│   ├── portfolio_positioning.md
│   ├── project_brief.md
│   └── qa_checklist.md
├── outputs/
│   ├── figures/
│   │   ├── 01_user_funnel.png
│   │   ├── 02_weekly_active_users.png
│   │   ├── 03_cohort_retention_heatmap.png
│   │   ├── 04_product_views_vs_conversion.png
│   │   ├── 05_category_performance.png
│   │   ├── 06_checkout_abandonment_by_device.png
│   │   ├── 07_retention_by_acquisition_channel.png
│   │   ├── 08_cart_abandonment_by_category.png
│   │   └── 09_sql_shell_output.png
│   └── results/            (41 query outputs + _run_log.csv)
├── reports/
│   ├── Ecommerce_Product_Analytics_Analyst_Report.docx
│   └── Ecommerce_Product_Analytics_Analyst_Report.pdf
├── scripts/
│   ├── make_charts.py
│   └── run_sql_project.py
├── sql/
│   └── ecommerce_product_analytics.sql
├── .gitignore
├── README.md
└── requirements.txt
```